DFMA of the Geometric Compass
A Comprehensive Study on Improving Manufacturing and Usability



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1.Problem Statement

The geometric compass, a fundamental tool in drafting and design, has long suffered from inefficiencies in its construction. Current designs often include redundant components that escalate manufacturing costs, complicate assembly processes, and reduce overall operational efficiency. These inefficiencies hinder standardization, increase production variability, and limit the potential for scalable improvements. Therefore, redesigning the compass to address these inefficiencies is crucial for enhancing functionality, reducing costs, and improving user satisfaction.

2.Objectives

1. **Optimize Design**: Develop a simplified compass design by minimizing part count and interfaces while maintaining all necessary functionalities.
2. **Enhance Efficiency**: Increase Theoretical Part Count Efficiency (TPCE) by aligning actual part count with theoretical minimums.
3. **Promote Standardization**: Identify components that can be standardized to reduce variability and simplify production.
4. **Streamline Assembly**: Simplify the assembly process to lower cycle times and minimize errors.
5. **Reduce Costs**: Eliminate unnecessary components to lower manufacturing and operational expenses.

3. Old Design

3.1 Introduction

The initial geometric compass design, while functional, includes unnecessary parts and interfaces that increase complexity and costs. The old design's inefficiencies make it less suited for scalable production and user-friendly operation.

3.2 Breakdown of Compass Parts

The compass consists of the following essential components:

1. Legs
2. Needle Point
3. Pencil Holder
4. Screw Knob
5. Joint or Pivot Mechanism
6. Knurled Screw
7. Hinge or Pivot Housing
8. Thumb Knob





3.3 Functional Analysis

Each part contributes to the functional integrity of the compass. The following table summarizes the components, number of parts (Np), and interfaces (Ni) required for functionality.

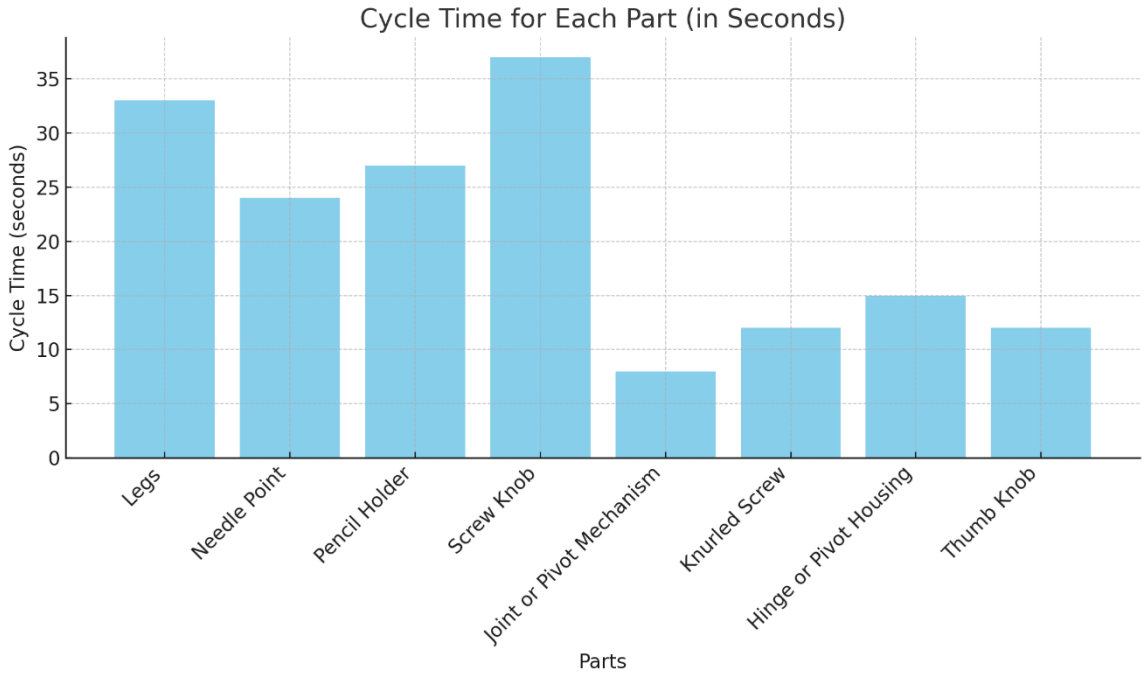
Part No.	Part	Np	Ni	Can Be Standardized	Practical Minimum Part
1.1	Legs	2	2	Yes	Yes
1.2	Needle Point	1	1	Yes	Yes
1.3	Pencil Holder	1	1	Yes	Yes
1.4	Screw Knob	1	1	Yes	Yes
1.5	Joint or Pivot Mechanism	1	3	No	No
1.6	Knurled Screw	1	1	Yes	No
1.7	Hinge or Pivot Housing	1	2	No	Yes
1.8	Thumb Knob	1	2	Yes	No

3.4 EXCEL SHEET OF COMPASS DFA

Part		*DFA Complexity*		*Functional Analysis/ Redesign opportunity*				Error Proofing		Handling			Insertion				Secondary operations				
Part Number	GEOMETRY COMPASS	*Numbers of parts*	*Number of interfaces (Ni) Part a to part b = 1*	*Theoretical Minimum Part Functional Analysis Chart*	*Part can be standardized (if not already standard)*	*Cost (Low, Medium*	*Practical Minimum*	Assumable wrong part (Gauge part)	Assumable part wrong way around	Tangle (Neat, Stack, Together)	Flexible (Fragile, Sharp, Slippery)	Pliers/ Tweezers/ Magnet/ peg plate	Difficult to align/ Locate	Holding down required	Resistance to insertion	Obstructed access/ visibility	Re-oriented work piece	Stress (Drill/ Twist/ Bend/ Bend/ Crimp)	Weld/ Solder/ Glue	Paint/ Label/ Heat/ Apply liquid or gas	Test (Measure/ Adjust)
1																					
1.1	*Legs*	*2*	*2*	*Y*	*Y*	*L*	*Y*	Y	Y	N	Y	Y	Y	Y	Y	Y	Y	N	N	N	Y
1.2	*Needle Point*	*1*	*1*	*Y*	*Y*	*L*	*Y*	N	N	N	N	Y	Y	Y	Y	Y	Y	Y	N	N	Y
1.3	*Pencil Holder*	*1*	*1*	*Y*	*Y*	*L*	*Y*	Y	Y	N	N	N	Y	Y	N	Y	Y	N	N	N	Y
1.4	*Screw Knob*	*1*	*1*	*Y*	*Y*	*L*	*Y*	N	N	N	N	N	Y	Y	Y	Y	Y	Y	Y	N	Y
1.5	*Joint*	*1*	*3*	*Y*	*N*	*L*	*N*	Y	Y	N	N	N	N	N	N	N	N	N	N	N	Y
1.6	*Knurled Screw*	*1*	*1*	*N*	*Y*	*L*	*N*	Y	Y	N	N	N	N	N	N	N	N	N	N	N	Y
1.7	*Hinge or Pivot Mecha*	*1*	*2*	*N*	*N*	*L*	*Y*	Y	Y	N	N	N	N	N	N	N	N	N	N	N	Y
1.8	*Thumb Knob*	*1*	*2*	*N*	*Y*	*L*	*N*	Y	Y	Y	N	N	N	N	N	Y	Y	N	Y	N	Y
TOTAL		*9*	*13*	*5*	*6*		*5*	6	6	1	1	2	4	4	3	5	5	2	2	0	8

3.5 Cycle time

Part No.	Part	Cycle Time (sec)
1.1	Legs	33
1.2	Needle Point	24
1.3	Pencil Holder	27
1.4	Screw Knob	37
1.5	Joint or Pivot Mechanism	8
1.6	Knurled Screw	12
1.7	Hinge or Pivot Housing	15
1.8	Thumb Knob	12



3.6 Efficiency Analysis of the Geometric Compass Design

The geometric compass is a highly efficient design, balancing functionality with minimal redundancy. It comprises nine essential components: legs, needle point, pencil holder, joint or pivot mechanism, screw knob, knurled screw, hinge or pivot housing, and thumb knob. Each part interacts through strategic interfaces, such as the legs connecting to the joint and tool holders, ensuring precise and stable operation. The legs, for instance, feature two interfaces each, connecting them to the joint, needle point, and pencil holder, while the joint serves as the rotational hub with three interfaces, ensuring flexibility and adjustment precision.

3.7 Optimization and Standardization Opportunities

The theoretical minimum part count for the compass is six, encompassing the legs, needle point, pencil holder, joint, and screw knob. With an actual part count of nine, the Theoretical Part Count Efficiency (TPCE) is 66.67%, exceeding the target efficiency of 60%. This score reflects a well-optimized design, with limited redundancy. Further optimization could involve integrating components like the hinge, joint, and screw knob into a single assembly to simplify the structure and reduce production costs.

Standardization of key parts—such as legs, needle point, pencil holder, and screw knobs—across compass models would streamline manufacturing and enhance interchangeability. Additionally, ergonomic designs for components like the thumb knob could improve usability without increasing complexity. Simplifying interfaces and focusing on modular assembly would also reduce errors and enhance overall production efficiency.

In conclusion, the geometric compass exemplifies a well-balanced design, meeting its functional requirements with a high level of efficiency. While already surpassing target benchmarks, minor improvements in standardization and component integration could further enhance its design, making it even more cost-effective and practical.

4. Improved design of Compass Parts

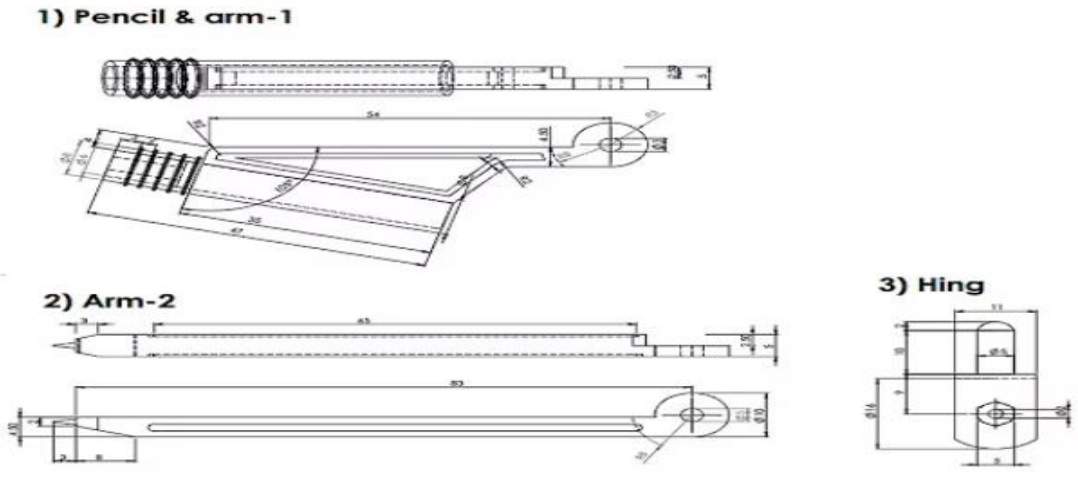
4.1 Introduction

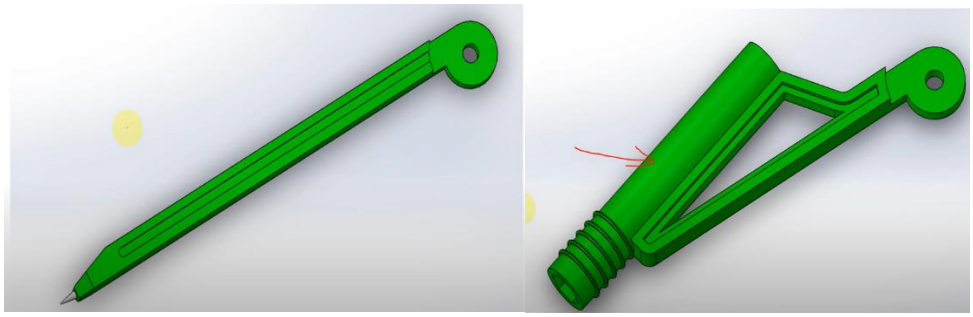
This report provides a detailed analysis of the parts of a geometric compass, including their functions, number of interfaces, and the potential for standardization. The goal is to evaluate the design and propose improvements for efficiency and manufacturability.

4.2 Breakdown of Compass Parts

The compass consists of the following visible and essential components:

1. Legs
2. Needle Point
3. Pencil Holder
4. Joint or Pivot Mechanism











4.3 Part Analysis Table

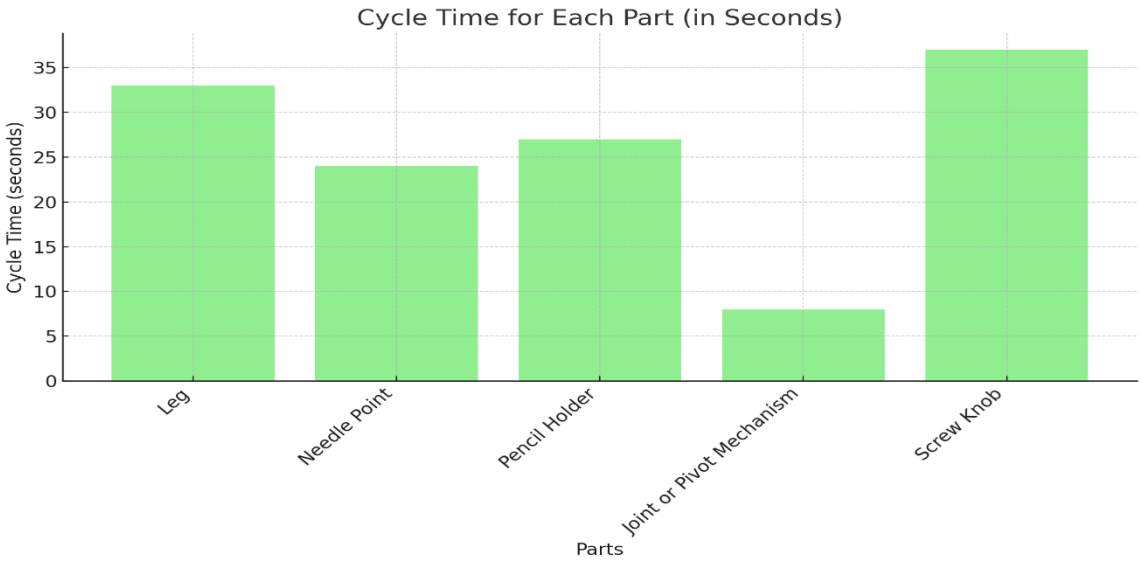
The following table summarizes the key aspects of the compass components, including the number of parts (Np), number of interfaces (Ni), potential for standardization, and practical importance.

Part No.	Part	Np (Number of Parts)	Ni (Number of Interfaces)	Standardization Potential
1.1	Leg	2	2	High
1.2	Needle Point	1	1	High
1.3	Pencil Holder	1	1	High
1.4	Joint or Pivot Mechanism	1	3	Moderate
1.5	Screw Knob	1		

Part		DFA Complexity		Functional Analysis/ Redesign opportunity				Error Proofing		Handling			Insertion				Secondary operations				
Part Number	Part Name	Numbers of parts (No)	Number of interfaces (Ni) Part a to part b = 1	Theoretical Minimum Part Functional Analysis Chart	Part can be standardized (if not already standard)	Cost (Low, Medium, High)	Practical Minimum Part	Assemble wrong part (Omit part)	Assemble part wrong way around	Tangle (Nest, Stick, Together)	Flexible (Fragile, Sharp, Slippery)	Pliers/ Tweezers/ Magnifying glass	Difficult to align/ Locate	Holding down required	Resistance to insertion	Obstructed access/ visibility	Re-oriented work piece	Screw (Drill/ Twist/ Rivet/ Bend/ Clamp)	Weld/ Solder/ Glue	Paint/ Lube/ Heat/ Apply liquid or gas	Test (Measure/ Adjust)
1																					
1.1	Leg	2	2	Y	Y	L	Y	Y	Y	N	Y	Y	Y	Y	Y	Y	Y	N	N	N	Y
1.2	Needle Point	1	1	Y	Y	L	Y	N	N	N	N	Y	Y	Y	Y	Y	Y	Y	N	N	Y
1.3	Pencil Holder	1	1	Y	N	L	N	Y	Y	N	N	N	Y	Y	N	Y	Y	N	N	N	Y
1.4	Joint or Pivot Mechanism	1	3	N	N	L	Y	Y	Y	N	N	N	N	N	N	N	N	N	N	N	Y
1.5	Screw Knob	1	1	Y	Y	L	Y	N	N	N	N	N	Y	Y	Y	Y	Y	Y	Y	N	Y
TOTAL		6	8	4	3		4	3	3	0	1	2	4	4	3	4	4	2	1	0	5

4.4 Cycle time

Part	Cycle Time (seconds)
Leg	33
Needle Point	24
Pencil Holder	27
Joint or Pivot Mechanism	8
Screw Knob	37



4.5 Observations and Analysis

The geometric compass design demonstrates an efficient and well-structured approach, with each part serving a vital function. The legs act as critical structural components, connecting to the joint and either the needle point or pencil holder. This ensures the compass remains stable and functional. The needle point

anchors the compass at a fixed position, interfacing with one leg, while the pencil holder securely houses the drawing tool and connects to the other leg. At the core of the design is the joint or pivot mechanism, which allows smooth rotation and precise adjustments by interfacing with both legs and the adjustment mechanisms.

4.6 Standardization Opportunities

Key components such as the legs, needle point, and pencil holder offer high potential for standardization. Their designs can be made uniform across different compass models, enabling cost-effective manufacturing and easier part replacement. The joint or pivot mechanism, while more specific in design, holds moderate standardization potential, depending on the type of compass and its intended use. These efforts could significantly improve production efficiency and reduce complexity in manufacturing processes.

4.7 Theoretical Part Count Efficiency (TPCE)

The Theoretical Part Count Efficiency (TPCE) is a measure of design efficiency, calculated as the ratio of the theoretical minimum number of parts to the total number of parts in a system, expressed as a percentage. A higher TPCE indicates a more optimized design with minimal redundancy.

Formula

The TPCE is calculated using the formula:

TPCE = (Theoretical Minimum Parts / Total Number of Parts) × 100

Data

Based on the provided table:

- Theoretical Minimum Parts = 4
- Total Number of Parts = 6

Calculation

Substituting the values into the formula:

TPCE = (4 / 6) × 100 = 66.67%

5. Comparison of Old and Improved Compass Designs

The redesign of the geometric compass emphasizes efficiency and practicality, addressing shortcomings in the old design while maintaining functionality.

5.1 Observations on the Old Design

The old design comprises nine components, including parts such as the knurled screw and thumb knob, which add unnecessary complexity. Additionally, the inclusion of the hinge housing increases assembly intricacies and manufacturing costs. With 13 interfaces, the design involves numerous interaction points, making assembly more challenging and increasing the likelihood of errors. Non-standardized parts like the joint and hinge housing further complicate the production process.

While the Theoretical Part Count Efficiency (TPCE) is 66.67% (calculated using the theoretical minimum of six parts versus the actual nine), this score meets the efficiency threshold but highlights room for

improvement. The redundant components, such as the knurled screw and thumb knob, contribute little to core functionality and inflate the part count.

5.2 Improvements in the New Design

The improved design reduces the total number of parts to six, focusing on essential components: the legs, needle point, pencil holder, joint, and screw knob. This streamlined structure eliminates non-essential parts, including the thumb knob and knurled screw, which simplifies manufacturing and assembly.
The reduction in interfaces to eight further enhances the ease of assembly and minimizes potential failure points. The new design also prioritizes standardization, particularly for components like the legs, needle point, and pencil holder. Standardized components lower production costs and allow for easier interchangeability across models.
Despite retaining the same TPCE score of 66.67% (four theoretical parts versus six total), the revised design demonstrates a better focus on core functionality and practicality.

5.3 Key Improvements in the New Design

1. **Reduction in Parts**:
   The elimination of redundant components like the thumb knob and knurled screw has reduced part count and improved design efficiency.
2. **Simplified Assembly**:
   Fewer interfaces result in a more straightforward assembly process, decreasing both manufacturing time and associated labor costs.
3. **Increased Standardization**:
   High potential for standardizing critical components across compass models ensures better consistency and cost savings during production.
4. **Optimized Functionality**:
   The new design prioritizes essential functions, aligning with principles of design efficiency while minimizing redundancy.

6. Cycle Time Comparison and Efficiency Analysis

6.1 Cycle Time Comparison

Part	Initial Cycle Time (sec)	New Cycle Time (sec)	Difference (sec)
Leg	33	33.0	0.0
Needle Point	24	24.0	0.0
Pencil Holder	27	27.0	0.0
Joint or Pivot Mechanism	8	8.0	0.0
Screw Knob	37	37.0	0.0
Knurled Screw	12	Removed	N/A
Hinge or Pivot Housing	15	Removed	N/A
Thumb Knob	12	Removed	N/A

6.2 Efficiency Summary
Initial Total Cycle Time (sec): 168
New Total Cycle Time (sec): 129
Efficiency Improvement (%): 23.21

7. Conclusion

The redesign of the geometric compass demonstrates significant advancements in efficiency, manufacturability, and functionality. By reducing the part count from nine to six and eliminating redundant components such as the knurled screw, thumb knob, and hinge housing, the new design streamlines the assembly process and lowers manufacturing costs. Standardizing critical parts like the legs, needle point, and pencil holder further enhances scalability and interchangeability across models. The revised design maintains the same Theoretical Part Count Efficiency (TPCE) of 66.67% but achieves a 23.21% reduction in total cycle time, underscoring its improved practicality. By addressing inefficiencies in the old design, the optimized compass ensures higher reliability, easier production, and a more user-friendly experience, setting a new standard for drafting tools.